

33. Use resolution to show that the compound proposition $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ is not satisfiable.

Soln: Let, the given compound proposition be satisfiable. $\therefore$ By defn. of satisfiability, there exists an assignment of truth values to p and q such that $(p \vee q) \wedge (\neg p \vee q) \wedge (p \vee \neg q) \wedge (\neg p \vee \neg q)$ is True. By defn. of AND, each of $(p \vee q)$, $(\neg p \vee q)$, $(p \vee \neg q)$ and $(\neg p \vee \neg q)$ are respectively True.

Clause $C_1: (p \vee q)$ Clause $C_2: (\neg p \vee q)$ Clause $C_3: (p \vee \neg q)$ Clause $C_4: (\neg p \vee \neg q)$

By Resolution on C_1, C_2 we get $C_5: (q \vee q)$

By Resolution on C_3, C_4 we get $C_6: (\neg q \vee \neg q)$

By Resolution on C_5, C_6 we get $C_7: \text{contradiction} \square$

~~By Resolution on C_1, C_3 we get $C_8: p$~~

$\therefore$ The compound proposition is not satisfiable.

34. The Logic Problem, taken from WFF'N PROOF, The Game of Logic, has these two assumptions:

1. "Logic is difficult or not many students like logic."
2. "If mathematics is easy, then logic is not difficult."

By translating these assumptions to statements involving propositional variables and logical connectives, determine whether each of the following are valid conclusions of these assumptions:

- p : Logic is difficult q : Many students like Logic
 r : Mathematics is easy.

Premises: 1. $p \vee \neg q$ 2. $r \rightarrow \neg p$

c) That mathematics is not easy, if many students like logic.